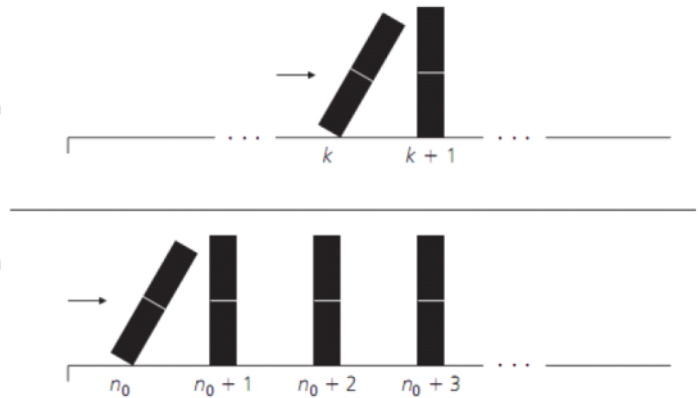

COMP 2121
DISCRETE
MATHEMATICS

Lecture 13



Mathematical Induction

- A2 is due Nov 5 in the lecture (paper work)
- Quiz 4 in the lab next week
based on Lab 7

Proof Outline

Mathematical induction is one of the more recently developed techniques of proof in the history of mathematics (used by Francesco Maurolico – 1575, Pierre de Fermat and Blaise Pascal and defined by Augustus De Morgan in 1883).

In natural sciences, deduction and induction are presented as alternative modes of thought: *deduction* being to infer a conclusion from general principles, *induction* being to enunciate a general principle after observing it to hold in a large number of specific instances.

Mathematical induction is a mathematical method to prove that a certain principle holds in processes that occur repeatedly and according to definite patterns.

PRINCIPLE OF MATHEMATICAL INDUCTION. Let $S(n)$ be a statement that is defined for integers n , and let a be a fixed integer. Suppose that the following two statements are true:

1. $S(a)$ is true
2. For all integers $k \geq a$, if $S(k)$ is true, then $S(k+1)$ is true.

Then, the statement “For all integers $n \geq a$, $S(n)$ ” is true.

Hence, proving a statement by mathematical induction is a **two-step process**:

- ✓ Step 1 is named the **basic step**. In step 1 we verify $S(a)$ is true.
- ✓ Step 2 is named the **inductive step**. First we assume that $S(k)$ is true (we call this *inductive hypothesis*). Then we prove that $S(k+1)$ is true.

Principle of Mathematical Induction Stated Formally is:

$$[S(a) \wedge \forall k \geq a (S(k) \rightarrow S(k+1))] \rightarrow \forall n \geq a (S(n))$$

$$S(1)$$

$$S(1) \rightarrow S(2)$$

$$S(2) \rightarrow S(3)$$

:

$$S(99) \rightarrow S(100)$$

$$S(100)$$



$$S(1)$$

$$S(k) \rightarrow S(k+1) \quad ; \quad \forall k \geq 1$$

$$\therefore S(n)$$

or

$$a \rightarrow b$$

$$\therefore b$$

Summation

prove

Example 1. Sum of the first n integers.

For all $n \geq 1$,

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

① Base case, verifying true for $n=1$

$$LHS = 1$$

$$RHS = \frac{1(1+1)}{2} = 1 \quad \checkmark$$

$$\frac{3}{7} + \frac{2}{7} \cdot 7$$

② Assume the following is true for $n=k$

$$1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$$

← use this

③ Target for $n=k+1$

$$1 + 2 + 3 + \dots + (k+1) = \frac{(k+1)(k+2)}{2}$$

← prove

④ Proof

$$1 + 2 + \dots + (k+1) = \dots = \frac{(k+1)(k+2)}{2}$$

$$1 + 2 + 3 + \dots + (k+1) = \boxed{1 + 2 + 3 + \dots + k} + (k+1) = \frac{k(k+1)}{2} + \frac{(k+1)}{1} \cdot \frac{2}{2}$$

step 2

$$= \frac{k(k+1) + 2(k+1)}{2} = \frac{k^2 + k + 2k + 2}{2} = \frac{k^2 + 3k + 2}{2} = \frac{(k+1)(k+2)}{2} \quad \checkmark$$

↑
matches the target

Divisibility

Mathematical induction can be used to prove more than sequences. Divisibility, program correctness, number of steps in algorithm can be proved using mathematical induction.

Example 2. For all integers $n \geq 1$, $5^n + 2^{n+1} = 3q$, for some integer q .

Note: This means that $5^n + 2^{n+1}$ is divisible by 3.

n	$5^n + 2^{n+1}$
1	$5^1 + 2^{1+1} = 9$
2	$5^2 + 2^{2+1} = 25 + 8 = 33$
3	$5^3 + 2^{3+1} = 125$

① Base case, verify true for $n=1$

$$5^1 + 2^{1+1} = 5 + 4 = 9 = 3 \cdot 3 \quad (q=3)$$

② Assume the following is true for $n=k$

$$5^k + 2^{k+1} = 3q_1; q_1 \in \mathbb{Z}$$

← use this!

$$\boxed{5^k} = 3q_1 - 2^{k+1}$$

③ Target for $n=k+1$

$$5^{k+1} + 2^{k+2} = 3q_2; q_2 \in \mathbb{Z}$$

← can not use!
need to prove

④ Proof

$$5^{k+1} + 2^{k+2} = \dots \dots \dots = 3q_2; q_2 \in \mathbb{Z}$$

$$5^{k+1} + 2^{k+2} = 5 \cdot \boxed{5^k} + 2 \cdot 2^{k+1} = 5 \cdot (3q_1 - 2^{k+1}) + 2 \cdot 2^{k+1}$$

step 2

$$= 15q_1 - 5 \cdot 2^{k+1} + 2 \cdot 2^{k+1}$$

$$= 15q_1 - 3 \cdot 2^{k+1} = 3(5q_1 - 2^{k+1})$$

$$= \underbrace{3(5q_1 - 2^{k+1})}_{= q_2 \in \mathbb{Z}}$$

Example 3. What does the following function return? Can you prove that by mathematical induction?

MysteryFunction (positive integer n)

```

i = 1
j = 4
while i ≠ n do
    j = j + 2i + 3
    i = i + 1
end while
return j
    
```

n	code Func returns	Formula $j_n = (n+1)^2$
1	4	$(1+1)^2 = 2^2 = 4$
2	$4 + 2(1) + 3 = 9$	$(2+1)^2 = 3^2 = 9$
3	$9 + 2(2) + 3 = 16$	$(3+1)^2 = 4^2 = 16$

The function returns $j_n = (n+1)^2$; $\forall n \geq 1$

① Base case, verify true for $n=1$

code : when $n=1$ then $i=1, j=4$. while loop does not run. function returns 4

formula : $j_1 = (1+1)^2 = 2^2 = 4$ } verified.

② Assume the following is true for $n=k$

The function returns $j_k = (k+1)^2$

③ Target for $n=k+1$

The function returns $j_{k+1} = (k+1+1)^2 = (k+2)^2$

④ Proof

$n=k+1$ the code returns $\dots$ use step 2 use code $= (k+2)^2$

from the code $j_{k+1} = j_k + 2i + 3$
 $\downarrow$ step 2

$j_{k+1} = (k+1)^2 + 2i + 3$
 $\downarrow$ $i = k$ (not)

$$j_{k+1} = (k+1)^2 + 2i + 3$$

↑ in the last
run $i = k$

$$\begin{aligned}
 j_{k+1} &= (k+1)^2 + 2k + 3 \\
 &= \underbrace{(k+1)(k+1)} + 2k + 3 \\
 &= k^2 + k + k + 1 + 2k + 3 \\
 &= k^2 + 4k + 4 = (k+2)(k+2) \\
 &= (k+2)^2 \quad \checkmark
 \end{aligned}$$